PERSONAL DATA

Place, Date of Birth, Nationality: Vilvorde, Belgium / 11 June 1985 / Belgian
Institution : Université Libre de Bruxelles,
Address: Laboratoire G-Time, Department of Geosciences, Environment and Society, Université Libre de Bruxelles, Av. Franklin Roosevelt 50 – CP160/02, 1050 Brussels, Belgium
Personal address: 6, rue
Phone: +32485341293
Email: corentin.caudron@ulb.be

AREA OF EXPERTISE

My research focuses on volcanoes and more specifically seeks to understand magmatic hydrothermal systems (so-called 'wet volcanoes') using multi-disciplinary approaches such as continuous seismicity, (hydro-)acoustics and lake monitoring and degassing.

Keywords: Hydrothermal systems and gas-driven eruptions – Magma dynamics – Ambient seismic noise and tremor – Volcanic lakes – Infrasound – Hydroacoustics

APPOINTMENTS

2021-Today	**Associate Professor, ULB, Belgium**
2020-2022	**Research Assistant, INGV Bologna**
2018-2021	**Chargé de Recherche at Research Institute for Development (Institut de recherche pour le développement), ISTerre, Grenoble, France (permanent position)** *Main Project*: 'Transitions of volcano activity in volcano-hydrothermal hosted environments'
2017-2018	**Doctor-Assistant, University of Ghent, Belgium** *Main Project*: 'Long-term instabilities of volcano-hydrothermal systems'
2016-2017	**FNRS postdoctoral Fellow, Université Libre de Bruxelles, Belgium** *Main Project*: 'Towards Forecast Hydro-volcanic eruptions'
2015-2016	**Postdoctoral Fellow, University of Cambridge, UK** *Main Project*: 'Constrain the dynamics of 2010 Eyjafjallajökull and 2014-2015 Bárdabunga-Holuhraun eruptions using seismic noise'
2013-2015	**Postdoctoral Fellow, Earth Observatory of Singapore, Magma Transport Dynamics - Nanyang Technological University, Singapore** *Main Project* - 'Using an infrasonic array in Singapore to monitor volcanoes'

EDUCATION

2013	**Ph.D**. in Science, Université libre de Bruxelles (Belgium), Royal Observatory of Belgium

	Thesis: "Multi-disciplinary monitoring of Kawah Ijen volcano (East Java, Indonesia)" PhD advisors: Thierry Camelbeeck (ROB), Alain Bernard (ULB)
2009	**M.Sc**. in Earth sciences, Université Libre de Bruxelles (Belgium), Magna cum laude Master Thesis: « *Study of Sirung volcano-hydrothermal system (West Timor, Indonesia)*".
2007	**B.Sc.** in Earth Sciences, Université Libre de Bruxelles (Belgium), cum laude

SUPERVISION

Master Theses

1. Z. Spica: "H/V study of Kawah Ijen volcano" – May 2011 – Université Libre de Bruxelles.
2. G. van Kriekinge: "Volcano-seismic events location during the 2011-2012 crisis of Kawah Ijen volcano" – September 2013 – Université Libre de Bruxelles
3. M. Draps: "Study of Kawah Ijen caldera using satellite imagery products and seismic recordings" – June 2015 – Vrije Universiteit of Brussels
4. J. Subira: "Développement d'un outil de surveillance sismique en zone volcanique comme appui à la prévention des risques éruptifs" –September 2018 -Université de Liège
5. S. Nowe: "Unrest at Öræfajökull in 2017-2018 and its seismic signature" – June 2019 – Université Libre de Bruxelles / Belgium
6. T. Muzellec : 'Analyse et localisation des perturbations mécaniques dans la structure du volcan Kilauea pendant l'éruption majeur de 2018' - June 2019 – ISTerre, Grenoble / France
7. A. Verwimp: 'Linking degassing seeps with the shallow subsurface at the Laacher See volcanic lake (Eifel, Germany)' – UGhent, Belgium – September 2020
8. M. Braem : 'Étude du bruit et tremor sismique lors d'un déploiement dense de géophone et de fibre optique en contexte volcano-hydrothermal' – ULB 2023 – Master in Geology
9. H. De Lathauwer : 'Seismic noise associated with the Gunnuhver volcano-hydrothermal system' – UGhent 2023
   → best thesis award at Geologica Belgica
10. R. Vetri: "Denoising fiber optic data using deep neural networks" – ULB, polytech, 2023 – Master in Engineer
11. J. Govoorts: "Characterizing ice-rock-water interactions using seismic noise" – ULB 2023
    → 2nd best thesis award at Geologica Belgica – Master in Environment
12. A. Jguirim: "Surveillance du lac volcanique de Laach en Allemagne" – ULB 2023 – Master in Environment
13. N. Frenkel : 'Caractérisation de l'influence des fluides hydrothermaux et des propriétés du sous-sol sur la dynamique d'un système volcano-hydrothermal' – ULB 2023 – Master in Geology
14. Z. Chentouf: "Unsupervised deep learning classification of volcano geophysical signals" – ULB, polytech, 2022 (ongoing) – Master in Engineer

15. M. Vieira De Campos Rola Pereira : "On the use of passive techniques to monitor geothermal operation" – Master in Environment (2024)
16. C. Kouamou Djampo : 'Deep learning of fiber optic data' (2024) – Master in Engineer
17. V. Kozlov: 'Deep learning of volcano seismic data' (2024) – Master in Engineer
18. M. Thonar: 'Studying gas flow in Poás volcanic lake using passive acoustic gas flux inversion and spectrograms' (2024) – Master in Engineer
19. M. Gossez: 'Étude du système hydrothermal de l'Askja' (2025), ULB Master in Environment
20. D. Tchetmi Zouatom. Comparison of methods for seismic signal analysis: hierarchical clustering vs convolutional neural networks (June 2025), Master in Engineer – Informatics ULB
21. Ilias El Bannasri, Machine Learning to classify Volcano-seismic data (June 2025), Master in engineer, EN Energy, ULB/VUB

Internships

1. Arnaud Watlet (Ma1 student Université Libre de Bruxelles, Belgium): "Monitoring Kawah Ijen volcano using machine learning techniques" (Summer 2010, Royal Observatory of Belgium)
2. Zack Spica (Ma1 student Université Libre de Bruxelles, Belgium): "Volcanic activity of Kawah Ijen volcano using continuous techniques" (Summer 2010, Royal Observatory of Belgium)
3. Antoine Triantafyllou (Ma1 student Université Libre de Bruxelles, Belgium): "Analysis of lineaments of the Kawah Ijen caldera from satellite images" (Summer 2010, Royal Observatory of Belgium)
4. Patrizia Battisti (Ma1 student Université Libre de Bruxelles, Belgium): "Historical activity of Kawah Ijen volcano" (Fall 2011, Royal Observatory of Belgium)
5. Imam Catur Priambodo (CVGHM, Indonesia): "Analyzing continuous data prior to the 2007 Kelud eruption, Indonesia" (February- May 2013, Royal Observatory of Belgium)
6. Wening Sulistri (CVGHM, Indonesia): "Analyzing continuous data of Lokon volcano, Indonesia" (February- May 2013, Royal Observatory of Belgium)
7. Clarence Chan Kok Wei (Ba1 student Nanyang Technlogical University, Singapore): "Development of a rapid infrasound deployment" (June-October 2015, EOS, Singapore)
8. Lukas Preiswerk (PhD student ETH Zurich, Switzerland): "Noise-based analysis of the Birchgletsher glacier, Switzerland" (August 2016, Royal Observatory of Belgium)
9. Nicolas Compaire (Ba3 student University of Strasbourg, France): "Seismic velocity associated with gas-driven eruptions at Kawah Ijen volcano (East Java, Indonesia)" (2017, ROB, Belgium)
10. Alexander Wickham-Piotrowski (Ma Student): Searching for seasonal variations of seismic wave velocity in relation with changes of the level of aquifers by seismic noise correlation techniques (3rd year engineering geology course, ISTerre, Grenoble)
11. Jeanne Andre: Stage chez EERA : supervision académique (Bio-ingénieur ULB, 2023)
12. Sonia Heuninck : Towards predicting volcanic eruptions; Ms student from Lyon (3 months at ULB; 2024)
13. Hazal Bektas: Etna volcano monitoring using seismic noise; Ms student from Istanbul Technological University (July 2024)
14. Sonia Heuninck: Fiber Bragg Grating technology in volcanology; Ms student from Lyon (June 2025)

Co-Supervisor of 2 *qualification projects* for PhD candidates (June-November 2015, EOS, Singapore):

- D. Mele Veedu: “Investigation of a phreatic and a lava dome eruption at Mayon volcano using ambient seismic noise and tremor”, Earth Observatory of Singapore
- P. Nanjundiah: “Toward constraining the origin of seismic crises at Gamalama volcano (Indonesia) using ambient seismic noise and tremor”, Earth Observatory of Singapore

PhD informal (co-)supervisor

- C. Donaldson: “Seismic velocity variations at Kilauea volcano and during the 2014-2015 Bárdabunga-Holuhraun eruptions”, University of Cambridge (completed in 2019)
- R. De Plaen: “Single-station monitoring of volcanoes using seismic ambient noise”, University of Luxembourg” (completed in November 2017)

PhD formal (co-)supervisor

1. L. Vanhooren: “Continuous electrical resistivity monitoring of hydrothermal areas” (started in March 2022, co-supervision UGhent)
2. L. Brenot: “Real-time monitoring of Volcanic UnresT using machine learning of seIsmic and remOte seNsing data.” (FNRS Aspirant, started in October 2022 at ULB, co-supervision with U Fairbanks (Alaska, USA))
3. O. Fontaine: “Characterizing volcanic tremor using dense seismic deployments in Iceland” (Fria funding, started in October 2022 at ULB)
4. Martanto: “Vers une classification automatisée des évènements volcano-sismiques pour l’archipel indonésien“ (ULB-ARES, 4 years, started in Sept 2022 at ULB)
5. J. Govoorts: “Towards Improved understanding of subglacial hydrology using passive seismic-based approaches and fiber optics” (FNRS Aspirant, started in October 2023 at ULB in co-supervision with K. Van Noten (Royal Observatory of Belgium) and C. Caucheteur (UMons)
6. J. Patzel: “Towards monitoring geothermal operations in 4D using passive seismics” (Wel-T FNRS funding, started in November 2023)
7. H. Bakkar Hindeleh: “Modelling multiple scenarios at three active crater lakes combining seismo-acoustic data with a multidisciplinary approach.” (FNRS Aspirant, will start October 2024 at ULB, co-supervision with U Victoria (New Zealand))
8. P. Tribel: “Learning Operators for Geophysic Phenomena Spatio-Temporal Simulation and Forecasting” co-supervisor (started in Oct 2023 as teaching assistant in Informatics at ULB)
9. L. van der Laat: “Avances en la Comprensión de la Sismicidad y Acústica Volcánica Análisis Multiparamétricos y Nuevas Técnicas de Monitoreo” (startin in 2025

Completed

A. Yates: ‘Towards an understanding of magma transport and storage mechanisms before explosive eruption by seismic interferometry and study of deformation” (2020-2023, ISTerre, France); currently PDoc at ULB.

J. Subira: “Machine Learning classification of volcano-seismic events in the Virunga Volcanic Province“(2020-2024, ULiège), currently staff scientist at Goma Volcano Observatory

S. Nowé: “Characterization of cryoseismic and volcano seismic sources in the Vatnajokull icecap region“(2019-2024, ULB – co-supervision with Royal Observatory of Belgium) currently professor in secondary school and teaching assistant at ULB.

Postdoc formal supervisor

- B. Roche: “Hydroacoustic of volcanic signals” (October 2023 – present, FNRS MIS Funding)

- A. Yates: “Seismic interferometry in geothermal environments” (March 2024 - present, Wel-T Funding)
- A. Watlet: “Forest Pulse” (Sept 2025, Marie Curie Postdoctoral Fellowship, secondment at ULB)

Former Postdoc formal co-supervisor

- Y. Cubuk: “Real-time monitoring of volcanoes in Iceland using seismic velocity” (May 2019 -2024)
- J. Soubestre: “Monitoring of tremor in Iceland” (August 2021-May 2024)
- R. Sanderson: “Machine-Learning of tremor in Iceland” (December 2022-2024)

TEACHING

Ongoing:

Coordinator: Geophysics (since 2022-23, 5ECTS) – Bachelor in Geology
Coordinator: Georessources (since 2023-24, 5 ECTS) – Master in Environment
E. Belsack, A. Van Hecke, **C. Caudron**.
→ Analyse d'un dispositif d'évaluation ludique inspiré de la Fresque du Climat, Journées scientifiques Ludopédagogie, Rennes, 2025.
Coordinator: Applied Geodynamics (since 2023-24, 5 ECTS) – Master in Geology

Co-lecturer for Volcanology course (2021-22-23, 4 hrs lectures)
Co-lecturer: Ressources: Genèse et environnement (2021-22-23, 4 hrs)
Co-lecturer : cours transversal de durabilité (2022-23, 2 hrs)

Civis school: 3 hours in 2024 on Monitoring Volcanic eruptions

Past

Lecturer in charge of the course of Volcanology (UGhent, 2019, 60 hrs, with M-A del Marmol and invited lecturers)

Invited Lecturer for the course of Volcanology (UGhent, 2018, 8 hrs: ‘Volcano-hydrothermal systems and Volcanology’, Course Lecturer: M. Kervyn)

Co-lecturer of ‘Imaging techniques in consolidated and unconsolidated Sediments’ (UGhent-Ma1, 2018 / lecturer: Veerle Cnudde)

Invited Lecturer for the course ‘Géophysique et tectonophysique’ (ULB-BA3, 2018,2020,2021: 2 hrs : Course / lecturer: T. Lecocq)

Invited Lecturer for the courses ‘Géophysique et tectonophysique (ULB-BA3, 2017: 6 hrs : Course / lecturers: T. Camelbeeck and T. Lecocq) and ‘Volcanologie’ (ULB-MA1, 2017: 9 hrs / lecturer: A. Bernard)

Invited Lecture for the course ‘Introduction to volcano monitoring using recent technology and machine learning in geoscience’ (ITB, Bandung, July 2021, 2 hrs)

Supervision for various fieldworks in geophysics in Belgium and abroad

Trainings: Advanced workshop on PhD supervision / 7 training to improve teaching and PhD students supervision

TECHNIQUES/RESEARCH SKILLS/TRAINING

Fieldwork experience: seismic vault construction and seismic station installation (broadband, short-period and nodes) / seismic network planning-reconfiguration / large and short-aperture infrasound arrays installation / Seismic nodes and Distributed Acoustic Sensing deployments / geophysical prospecting (ERI, GPR, H/V, EM) / echo sounding and hydrophone installation / CO_2 measuring techniques / fluid sampling for geochemical analyses / Thermal IR camera / Instruments installation for lake monitoring (level, temperature,…) [Locations: Indonesia, Philippines, Costa Rica, Hawai'i, Germany, France, Italy,etc]

Geochemistry analytical experience: HPLC (ULB), SEM (ULB, UGhent), XRD (ULB), CT-scan (UGhent), spectrophotometry (ULB), pH/temperature/alkalinity measurements/sample preparation for sulphur isotopes

Data processing: seismic, infrasound, hydrophone, echo sounding data, time series analysis and satellite data, machine learning (supervised and unsupervised)

Modelling: Phreeqc, Geochemist Workbench, TOUGH2 (beginner)

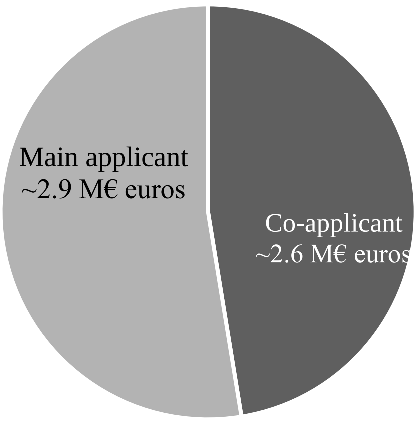
Programming: python environment (developer for MSNoise, good experience with Obspy toolbox, matplotlib, numpy, scipy), Matlab (limited experience)

Software: QGis, Global Mapper, Envi, CorelDraw, Microsoft Office suite, Agisoft, MSNoise, SARA, Tough2, Inferno, PMCC, Geopsy, SeisGram, Golden Surfer

Project writing: ERC, NERC, CNRS/IRD (France), ANR, volcano group projects at Earth Observatory of Singapore, Brains project (Belspo, Belgium)

Trainings: Geopsy course (Las Vegas – February 2012), SEG course(Paris - June 2012), Project Manager Formation (ALTRAN – Brussels – October 2012), Infrasound training at ISLA (Infrasound Laboratory University of Hawaii, Hawai'i, USA - February-March 2015), TOUGH2 (training at the University of Bologna, Italy - March 2017), DataCamp online training – Data Scientist and Machine-Learning

AWARDS / FUNDING / GRANTS



Awards

2010: Best presentation award: PhD Student day of ENVITAM school (Namur, Belgium): January 2010
2018: Prize Baron Van Ertborn (best publication in Geosciences) – Académie Royale des Sciences (Belgium)

2024: Prix de la diffusion scientifique, Lab Discovery Tour, ULB, 2024

Past funding

2009-2013: Action2 PhD grant (**PI**: 160 k€)
2013: Master thesis CUD fieldwork grant (3 k€)
2013: IAVCEI -Commission of Volcanic Lake travel grant (2 k€)
2014-2015: Infrasound and Lab Volcanoes research projects at the Earth Observatory of Singapore (Singapore) (**Co-Investigator**, project renewals/ PI : Benoît Taisne).
2015-2016: Fondation Wiener Anspach Postdoctoral Grant (**PI**: 52 k€)
2016-2019 : FNRS Chargé de Recherches Postdoctoral Grant (**PI**; 177 k€)
2017: NERC Advanced Training Short Course (UK) grant (**Co-PI** - PI: S. De Angelis; 14 k€)
2016-2019: Lab volcanoes project (**Research Partner** – PI: Benoît Taisne, EOS-NTU, Singapore; 250 k€)
2019: *Interactions between eruptive activity and sector collapse at Anak Krakatau, Indonesia* (**Research Partner** – NERC / PIs: Sebastian Watt, Mike Cassidy, Samantha Engwell, Mike Burton, 50 k€)
2018-2021: *IS-Noise*: Using Seismic Noise to Monitor Iceland (**Research Partner** – PI: Kristín Jónsdóttir, IMO, Iceland – Icelandic Research Fund, collaborators: Royal Observatory of Belgium, Uppsala University (Sweden); 410 k€)
2019-2021: *Towards Real-time Acoustic monitoring of volcanic gas emissions in aqueous environment* (To ReAct), **PI** (BQR; 15 k€)
2019: FNRS FRIA PhD fellowship of S. Nowé: "Characterization of cryoseismic and volcano seismic sources in the Vatnajokull icecap region"

2020-2021: *Quantification acoustique du dégazage* (**co-PI**; PI: J. Vandemeulebrouck; 6 k€) – AAP USMB, France.
2020-2023: *Application of Artificial Intelligence for understanding and prediction of human and volcanic activity* (**Research Partner** – PI: E. Trouvé; 180 k€) – ANR, collaborators: LISTIC (France)
2020-2021 : *Validation d'une approche hydroacoustique pour quantifier les flux de méthane transférés du compartiment benthique d'un écosystème lacustre à l'atmosphère par bullage* (**Research Partner** – PI: E. Lyautey ; 8 k€) – AAP Research, collaborators: ISTerre, CARRTEL
2020-2021: *Geodetic and hydrological controls on seismic velocity changes after large earthquakes* (**Research Partner** – PI: M. Savage; 60 k€) – EQC New Zealand, collaborators: GNS Science, U Wellington
2020-2024 Quantifying underwater volcano degassing using novel seismo-acoustic approaches (**co-PI** / PI: Zack Spica; 20 k$) – Thomas Jefferson Fund – collaborator: U Michigan (US)
2021: *Vers une meilleure compréhension des variations de vitesse sismique en milieu volcanique* (**PI**) – PHC Tournesol, France, collaborator: T. Lecocq, 2.4k€)
2021: *Towards forecasting eruptions at Indonesian volcanoes using machine learning of volcano seismic data* (**PI**), Campus France, Nusantara (10 k€)
2021-23: *Passive Volcano degassing Quantification in aquatic systems* (**PI** ; UGA young researcher 14 k€)
2022: ULB Research Department Incentive «Characterizing volcanic tremor using dense seismic deployments in Iceland » for O. Fontaine
2023: LabDiscovery Tour: The Geoscience Trail Innoviris (**PI** ; 10 k€)
2021-2024: Seismic tremor for real-time tracking of Natural Hazards in Iceland (**co-PI** – PI: K. Jónsdóttir (IMO, Iceland) and T. Lecocq (ROB, Belgium); 950 k€)

<u>Ongoing</u>

As co(-PI)

2022: FNRS Aspirant PhD fellowship of L. Brenot: "Real-time monitoring of Volcanic UnresT using machine learning of seIsmic and remOte seNsing data."

2022: FNRS FRIA PhD fellowship of O. Fontaine: "Characterizing volcanic tremor using dense seismic deployments in Iceland"

2022: ARES PhD fellowship of Martanto: "Vers une classification automatisée des évènements volcano-sismiques pour l'archipel indonésien"

2023: FNRS Aspirant PhD fellowship of J. Govoorts: "Towards Improved understanding of subglacial hydrology using passive seismic-based approaches and fiber optics" (FNRS Aspirant, started in October 2023 at ULB in co-supervision with K. Van Noten (Royal Observatory of Belgium) and C. Caucheteur (UMons)

2024: FNRS Aspirant PhD fellowship of H. Bakkar: "Modelling multiple scenarios at three active crater lakes combining seismo-acoustic data with a multidisciplinary approach." (FNRS Aspirant, started in October 2024 at ULB in co-supervision with F. Illsley-Kemp (U Victoria, New Zealand)

2022-2026: Can ERT reveal the dynamics of volcanic hydrothermal systems? (ERupT, **co-PI**, PI : T. Hermans (UGent, Belgium), FWO project (Belgium), ~450 k€)

2023-2026: CalderaSounds MIS-FNRS Incentive grant for scientific research (Belgium) (**PI** ; 460 k€)

2023-2027: WEL-T Starting Grant Investigator Program (Belgium) *Geo4D: Towards 4D Monitoring of Geothermal Operations in 4D using passive-based seismic approaches* (**PI** ; 800 k€)

2023-2025: The seismic signature of caldera reinflation at Askja Volcano, Iceland (**PI** ; 30 k€; co-PIs: N. Rawlinson (U Cambridge) and T. Winder (U Iceland)) – Fondation Wiener Anspach Project

2024-2026: Monitoring underwater volcano degassing using novel hydrophone technology - Catalyst (PI: 45 keuros; co-PI Oliver Lamb (GNS, New Zealand)).

2024-2028: Surveillance du paysage sonore sous-marin pour comprendre les mécanismes responsables des regains d'activité des calderas volcaniques (PI : 200 keuros, ULB Arc Consolidator Funding)

As partner

2023-2028: Developing a new platform for assessing volcano hazard through analysis of globally acquired monitoring data (Lead-PI: B. Taisne ~S$3.1M funded by Singapore National Research Foundation)

COLLABORATION (active)

Europe: *University of Cambridge, UK*: N. Rawlinson, *University of Iceland, Iceland*: Tom Winder, *IMO Iceland:* K. Jónsdóttir, M. Pfeiffer, *GFZ*-Potsdam: P. Jousset, *LMU Munich, Germany*: Anthony Lamur, *U Liverpool, UK*: S. De Angelis, *ETH Zurich Switzerland*: A. Fichtner, *IPGP France:* L. Seydoux, *ISTerre France:* J. Vandemeulebrouck, V. Pinel *INGV-Bologna Italy:* D. Rouwet, L. Zaccarelli *, INGV-Catania Italy:* M. Longo, *University of Firenze, Italy*: F. Tassi, *ECGS, Luxembourg*: J. Barrière, *U Granada*: J. Prudencio, *U Mainz*: M. Reiss, *UGent:* M. De Batist, T. Hermans, *Royal Observatory of Belgium*: T. Lecocq, K. Van Noten, *ULiège:* F. Nguyen.

International: *Cornell,* USA: C. Munoz-Saez, *U Fairbanks Alaska, USA:* T. Girona*, CVGHM, Indonesia*: D. Syahbana, *ITB, Indonesia*: A.D. Nugraha, *GNS, New Zealand*: B. Christenson, O. Lamb, G. Kilgour, *University of Canterbury, New Zealand*: D. Dempsey, A. Ardid, *University of Wellington, New Zealand*: M. Savage, *UNAM, Mexico*: R. Campion, *Centro de Investigación Científica y de Educación Superior de Ensenada, Baja California*: L. Peiffer *OVSCICORI, Costa Rica*: M. de Moor, *EOS, Singapore*: B. Taisne, *Earthquake Research Institute, University of Tokyo, Japan*: Y. Aoki.

COMMUNITY CONTRIBUTIONS

International responsibilities

- SAC (Scientific Advisory Committee, funded by the British government) for Montserrat volcano (since 2024)
- Appointed member of the Collegium of the Belgian Academy (Classe Sciences, since 2026)
- Appointed effective member (IAVCEI) of the Belgian National
- Committee for Geodesy and Geophysics (BNCGG; since 2025)
- Co-chair of the IAVCEI/IASPEI Commission of Volcano Seismology and Acoustics (since 2026)
- I have also been selected to serve for the EPOS-France Scientific Committee for the Volcanology since 2024

Past

- Co-Leader of the IAVCEI Commission on Volcanic Lakes (https://iavcei-cvl.org/) (2019-2023); currently in the Steering Committee (2023-now).
- Chair of Focus Group (Volcanic Eruption Forecasting) on Artificial Intelligence for Natural Disaster Management (United Nations) (2021-2023)

More short-term expertise tasks are listed below:

- Merapi 2020 eruption & Nyiragongo 2021 eruptions task forces
- Évaluation Belge des Risques Nationaux – Centre de Crise National Belge : national expert for 'Volcanic Eruption Abroad' (2023)
- Campi Flegrei phreatic eruption SGR-CRV meeting (March 2024): invited to present precursors to phreatic eruption
- EarthScope PI Instrumentation Advisory Committee (2024)

At ULB

Vice and Président d'Inforsciences (2025-now) / Président of the Valorisation group of the CCMS (2026-now) / Membre du bureau de la Faculté des Sciences (2024-now)

Bureau of DGES (2022-2023) ; Secrétaire académique DGES (2023-2025) / Commission électorale Faculté des Sciences (2023-2025) / Festival des Sciences – intervenant / Data Ambassador at ULB / Uniter Doctoral School (Scientific Committee) / Groupe de Travail Open Science & construction des Unités d'Enseignement Sciences / Membre du jury de Ma Thèse en 180s (2025) / Perspectives Recherche with LaCambre and Ohme (2025 with G. Rolanda Ms student)

Alumni network of Fondation Wiener Anspach (since 2021)

Editor

Editor for Volcanica (2022-2024) and elected in the editorial board since 2024

Associate Editor for Journal of Geophysical Research-Solid Earth (2019-2021)

Guest Editor

Guest Invited Editor for a Special Volume 'Multiscale Approach and Integrated Multiplatform Techniques to Study the Volcanic and Geothermal Environments' (Frontiers in Earth Sciences)

Guest Invited Editor for a Special Volume 'The January 2022 Explosive Eruption of Hunga Volcano, Tonga' (Frontiers in Earth Sciences)

Editor IAVCEI Book series Advances in Volcanology and Active Volcanoes of the World: " "*Modern Volcano Monitoring*" (ongoing)

Guest Invited Editor for a Special Volume 'Towards Improved Forecasting of Volcanic Eruptions" (Frontiers in Earth Sciences, 2020)

Guest Invited Editor for a Special Volume "Geochemistry and Geophysics of Active Volcanic Lakes" (Geological Society of London, 2017)

Reviewer for International Journals

Nature Communications, Geophysical Research Letters, Journal of Geophysical Research, Geophysical Journal International, Earth Planet and Space, Frontiers in Earth Science, Journal of Volcanology and Geothermal Research, Seismological Research Letters, Pure and Applied Geophysics, Scientific Reports, Plos One, Journal of African Earth Sciences, Natural Hazards and Earth System Sciences, American Mineralogist, Bulletin of Seismological Society, NHESS, and different Special Volumes, Earth and Planetary Science Letters,etc… / since 2018 encoded in Publons

Reviewer for scientific proposals

ANR, France, 2017, 2024 / NSF, USA, 2019 / Fondecyt, Chile, 2019 / NERC, UK, 2022 / DFG Germany, 2023 / Czech Science Foundation 2024

Convener of scientific sessions

2013- COV8, Jogyakarta: Volcanic lakes and their impact on human and natural environments
2015–AGU (invited): Eruptive Processes and Watery Hazards of "Wet" Volcanoes on Land, in the Sea, or under Ice II

2015-AOGS (invited): State of the Art in Monitoring Volcanic Activities
2016-AGU (invited): New Methods to Forecast Volcanic Eruptions
2016-EGU (invited): Monitoring recent and Ongoing Volcanic Activity
2017-IAVCEI: Forecasting volcanic eruptions
2017-IAVCEI (invited): Wet volcanoes: aquifers and lakes and their related hazards
2018-COV10 (invited): Advanced and non-conventional seismic methods for monitoring active volcanoes
2019-IUGG (invited): Geochemistry and Geophysics of Active Crater Lakes
2019-IUGG (invited): Phreatic and Hydrothermal Eruptions: What We Really Know About Triggers, Magnitude, Styles and Hazards
2020-EGU (invited): Advances in monitoring techniques for challenging and extreme environments
2020 – COV10 (invited): What do volcano seismo-acoustic signals mean?
2020 – COV10 (invited): Progression of unrest in volcanic systems: An evaluation and a multiparameter update of the Generic Volcanic Earthquake Swarm Model (GVESM)
2021- IAVCEI: Monitoring and modelling active volcanic lakes
2021- IAVCEI (invited): Volcano seismology and acoustics: new approaches in monitoring and interpretation
2024 – EGU Galileo conference on Fiber Optics (Catania, Italy, June 2024) (invited)
2024 – Geologica Belgica Scientific Committee (Belgium, Sept 2024) (invited)
2025 – IAVCEI Gas-driven eruptions (Geneva, 2025) (invited)
2025 – IAVCEI Volcano seismology and acoustics (Geneva, 2025) (invited)

Examination and review

PhD committee

2019 : Raymond Massimo PhD Thesis – Université Libre de Bruxelles (Belgium)
2019 & 2020 : Julien Robic PhD Thesis – Université Libre de Bruxelles (Belgium)
2021: Roger Machacca PhD Thesis – ISTerre (France)
2022 to 2024 : Lorenzo Cappelli PhD Thesis – Université Libre de Bruxelles (Belgium)
2022-ongoing : Begra Disando Tabegra PhD Thesis – Université Libre de Bruxelles (Belgium)
2022-ongoing : Lorenzo Fabris PhD Thesis – Université Libre de Bruxelles (Belgium)
2024 : Théo Bouvard – U Namur (Belgium)

PhD jury

2019: Raymond Massimo PhD Thesis – Université Libre de Bruxelles (Belgium)
2022 : Sietze J. de Graaff PhD Thesis – Making Sense of Destruction, a Geochemical and Petrological Investigation of the Impact Melt Rocks and Crystalline Basement of the Chicxulub Impact Structure – VUB (Belgium) - secretary

2023: Amdemichael Zafu Tadesse - Magmatic Evolution and Tectonic Interactions of a Restless Volcanic Field and its Role in the Evolution of a Continental Rift in East Africa – Université Libre de Bruxelles (Belgium) – secretary of jury
2023: A. Iozzia: "Geophysical analyses and numerical modeling to characterize the eruptive and flank dynamics of Mt Etna" U Catania (Italy) – rapporteur

2024: J. Robic: “Isotopic characterization of active volcanic lakes, their associated magmatic-hydrothermal systems and high temperature fumaroles: from magma degassing to mineral precipitation” – ULB (Belgium) – president of jury
2024: P. Makus “Unravelling Contributions to the Dynamics of the Elastic Properties in Volcanic Environments” – Rapporteur – GFZ, Germany (November 2024)
2024: L. Pantobe “Identification et modélisation des processus physiques de déclenchement et modulation de la sismicité volcanique : applications à la Soufrière de Guadeloupe” – Examineur – Université Paris Cité (December 2024)

2025: J. Fone (U Cambridge, UK) “Ambient seismic noise imaging using trans-dimensional and machine learning techniques with application to Borneo and Iceland” - University of Cambridge (Jan 2025) - examiner

2025 : S. Klaasen (ETH, Zurich)

Ms Thesis jury

Since 2017, I have been in the jury of several Ms students for different master programs (geology, environment, engineers; ~5-15/year)

OUTREACH

Participation at Université des Enfants, Exhibitor at the Royal Society Exhibition (London, UK, 2016), Consultant for a National Geographic Explorer magazine, BBC science series ‘Forces of Nature’, Wired magazine, LiveScience, ULB research news-ULB Faculté des Sciences Research news-Objet de la Recherche, IRD, Daily Science, La Libre Belgique (2021 Canaries ; Enseignement Géologie, 2022), La Dernière Heure, Vers L’Avenir, Le Soir, Le Ligueur, RTBF TV-Radio (Les éclaireurs, Matin Première, 2020, Rtbf, CQFD, Declic, Journal Télévisé, Matin Première RTBF (23/11/2021), 2021 ; LaLibre (Tonga eruption ; Jan 2022), RTBF (Niouzz ; Jan 2022)), RTL, Le Labo des savoirs radio (Nantes, France), Pour la Science, New Scientist, LaCroix, FranceInfo (in 2020 and 2021), L’express, Reporterre, Sciences et Avenir, National Geographic, conferences in schools.

2021 : RTBF –Matin Première ; La Première 6-8 ; LaLibre ; National Geographic

2022 : Euradio Le siècle des ampoules : La surveillance des volcans; newsletters @ULB.

2023
Coffee Experts at ULB: fieldwork
March : lithotherapy « Le monde est Stone” Medor
Summer : Laacher See : LaLibre;; LaLibre et Metro;
October: Campi Flegrei: LaLibre
November – Iceland : RTL-TVI : JT 13h-19h – Focus 19h ; RTBF Déclic
December - Iceland :Lesoir : Eu!radio

2024
January: Eu!radio : geothermal energy, Iceland RTBF La Première
March : Iceland : RTBF
April: Brusolair Podcast / Fédération Indépendante des Seniors asbl
May: Journal des Enfants JDE

July – Italian volcanoes activity: BelRtl;
The Conversation France: Au cœur du volcan : mesurer les bulles d'un lac qui n'existe plus ou quand une expédition scientifique déraille ;
August : Ca m'intéresse / Prisma média : les Volcans Bleus ; RTBF : infographie volcans et nuage de soufre en Belgique

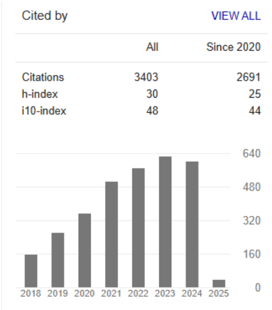
2025
January ; Vif Express : 'lithothérapie'
February: RTBF Matin Première Santorin / LeSoir Santorin / RTBF Quel Temps pour la Planète ? Les superéruptions volcaniques
March: Télépro 'Le monde de Jammy' Les superéruptions volcaniques
April : Radio Campus : AI to forecast eruptions
May : Etna eruption (Vivacité)
Juillet: Le Parisien / Kamchatka (Lalibre)
Aout: LaPremiere-Etna

LANGUAGES

French: mother tongue
English: fluent
Indonesian: good oral and reading skills
Russian: good (need practice)
Dutch: good (need practice)
Spanish: learning

PUBLICATIONS

Bibliometric indicators: Google scholar: h-index = 30– 3403 citations / Orcid



Peer-reviewed

As an editor:

Z. Spica, C. Caudron “Modern Volcano Monitoring”. IAVCEI Book series Advances in Volcanology and Active Volcanoes of the World, Springer Nature

D. Rouwet, F. Tassi, A. Mazot, C. Caudron. Volcanic Lake Dynamics and Related Hazards Frontiers in Earth Science (ongoing)

C. Caudron, Y. Aoki, N. Fournier, T. Girona, L. Chardot, Towards Improved Forecasting of Volcanic Eruptions, Frontiers in Earth Science

T. Ohba, B. Capaccioni and **C. Caudron**, Geochemistry and Geophysics of Active Volcanic Lakes, SP437, Geological Society of London, London, UK.

As an author:

*with PhD or Ms student

2025

[77]Lesage, P., **Caudron, C**. and Yates, A., 2025. Coda Wave Interferometry for Volcano Monitoring. In Modern Volcano Monitoring (pp. 147-188). Cham: Springer Nature Switzerland.

[76]Aiuppa, A., **Caudron, C.**, Chiodini, G., Ingebritsen, S. and Viveiros, F., 2025. Geochemical Monitoring of Volcanic Fluids in the Twenty-First Century. *Modern Volcano Monitoring*, pp.209-260.

[75]Jousset, P., Currenti, G., Murphy, S., Eibl, E.P., **Caudron, C.,** Retailleau, L., Wollin, C., Klaasen, S., Nishimura, T., Fichtner, A. and Spica, Z., 2025. Fiber optic sensing for volcano monitoring and imaging volcanic processes. In *Modern volcano monitoring* (pp. 509-567). Cham: Springer Nature Switzerland.

[74]Reiss, M.C., **Caudron, C.**, Hering, P. and Roman, D., 2025. Tremor signals reveal the structure and dynamics of the Oldoinyo Lengai magmatic plumbing system. Communications Earth & Environment, 6(1), p.781.

[73]Roche, B., **Caudron, C.,** Van der Laat, L., De Moor, J.M., Avard, G., Pacheco, J., Bakkar, H., Govoorts, J. and Rodriguez, A., 2025. First hydroacoustic recording of ebullition events in an active volcanic lake–Poás, Costa Rica. Volcanica, 8(1), pp.225-239.

[72]Yates, A., **Caudron, C**., Mordret, A., Lesage, P., Cannata, A., Cannavo, F., Lecocq, T., Pinel, V. and Zaccarelli, L., 2025. Precursory velocity changes prior to the 2019 paroxysms at Stromboli volcano, Italy, from coda wave interferometry. *Volcanica*, *8*(1), pp.203-223.

[71]Ardid, A., Dempsey, D., **Caudron, C**., Cronin, S., Kennedy, B., Girona, T., Roman, D., Miller, C., Potter, S., Lamb, O.D. and Martanto, A., 2025. Ergodic seismic precursors and transfer learning for short term eruption forecasting at data scarce volcanoes. *Nature communications*, *16*(1), p.1758.

[70]Soubestre, J., **Caudron, C**., Melnik, O., Lecocq, T., Jaupart, C., Shapiro, N.M., Journeau, C., Çubuk-Sabuncu, Y. and Jónsdóttir, K., 2025. Dynamics of the 2021 Fagradalsfjall eruption (Iceland) revealed by volcanic tremor patterns. *Journal of Geophysical Research: Solid Earth*, *130*(2), p.e2024JB029380.

2024

[69]Steinke, B., Jolly, A.D., Girona, T., **Caudron, C**., Bramwell, L.A., Cronin, S.J., Illsley-Kemp, F. and Hughes, E.C., 2024. Dynamics and Detection of Pulsed Tremor at Whakaari (White Island), Aotearoa New Zealand. *Geophysical Research Letters*, *51*(20), p.e2024GL110447.

[68]Svennevig, K., Hicks, S.P., Forbriger, T., Lecocq, T., Widmer-Schnidrig, R., Mangeney, A., Hibert, C., Korsgaard, N.J., Lucas, A., Satriano, C. and Anthony, R.E.,…, **C. Caudron**,… 2024. A rockslide-generated tsunami in a Greenland fjord rang Earth for 9 days. Science, 385(6714), pp.1196-1205.

[67]Yates, A.S.,* **Caudron, C.**, Mordret, A., Lesage, P., Pinel, V., Lecocq, T., Miller, C.A., Lamb, O.D. and Fournier, N., 2024. Seasonal snow cycles and their possible influence on seismic velocity changes and eruptive activity at Ruapehu volcano, New Zealand. *Journal of Geophysical Research: Solid Earth*, *129*(12), p.e2024JB029568.

[66] Robic, J. *, Debaille, V., Villacorte, E., Hublet, G., Maximo, R.P.R., Syahbana, D.K., **Caudron, C**., Bornas, M.A.V., Solidum Jr, R.U. and Bernard, A., 2024. Understanding zinc isotopic signatures in volcanic lakes. Geochimica et Cosmochimica Acta, 384, pp.178-193.

[65] van der Laat, L.*, Spica, Z., **Caudron, C**. and Girona, T., 2024. Magma fizz: Tremor during the Kīlauea summit reservoir decompression, Journal of Volcanology and Geothermal Research, p.108174.

[64]Çubuk-Sabuncu, Y., Jónsdóttir, K., Árnadóttir, T., Mordret, A., **Caudron, C.**, Lecocq, T. and De Plaen, R., 2024. Temporal seismic velocity changes associated with the Mw 6.1, May 2008 Ölfus doublet, South Iceland: A joint interpretation from dv/v and GPS. Journal of Geophysical Research: Solid Earth, 129(4), p.e2023JB027064.

[63] **Caudron, C.**, Miao, Y., Spica, Z.J., Wollin, C., Haberland, C., Jousset, P., Yates, A., Vandemeulebrouck, J., Schmidt, B., Krawczyk, C. and Dahm, T., 2024. Monitoring underwater volcano degassing using fiber-optic sensing. Scientific Reports, 14(1), p.3128.

[62] Contreras-Arratia, R., Chouet, B., **Caudron, C**., Nath, N., Balchan, A., Madoo, F. and Ramsingh, H., 2024. Tracking seismicity in an underfunded institution: The case of La Soufrière St Vincent volcanic eruption 2020–2021. Journal of Volcanology and Geothermal Research, p.107990.

2023

[61] Muzellec*, T., Lesage, P., **Caudron, C**. and Got, J.L., 2023. Migration of Mechanical Perturbations Estimated by Seismic Coda Wave Interferometry During the 2018 Pre-Eruptive Period at Kīlauea Volcano, Hawaii. Journal of Geophysical Research: Solid Earth, 128(8), p.e2022JB026224.

[60] Ardid, A., Dempsey, D., **Caudron, C.**, Cronin, S.J., Miller, C.A., Melchor, I., Syahbana, D. and Kennedy, B., Using template matching to detect hidden fluid release episodes beneath crater lakes in Ruapehu, Copahue and Kawah Ijen volcanoes. Journal of Geophysical Research: Solid Earth, p.e2023JB026729.

[59]**Caudron, C.** Vers une meilleure compréhension de la dynamique des systèmes volcano-hydrothermaux par la sismicité continue. Proceedings of the Royal Academy for Overseas Sciences, Belgium

[58] Perttu, A., Assink, J., Van Eaton, A.R., **Caudron, C**., Vagasky, C., Krippner, J., McKee, K., De Angelis, S., Perttu, B., Taisne, B. and Lube, G., 2023. Remote Characterization of the 12 January 2020 Eruption of Taal Volcano, Philippines, Using Seismo-Acoustic, Volcanic Lightning, and Satellite Observations. Bulletin of the Seismological Society of America, 113(4), pp.1471-1492.

[57]Machacca, R. *, Lesage, P., Tavera, H., Pesicek, J.D., **Caudron, C.**, Torres, J.L., Puma, N., Vargas, K., Lazarte, I., Rivera, M. and Burgisser, A., 2023. The 2013–2020 seismic activity at Sabancaya Volcano (Peru): Long lasting unrest and eruption. Journal of Volcanology and Geothermal Research, 435, p.107767.

[56]Subira, J. *, Barrière, J., **Caudron, C.**, Hubert-Ferrari, A., Oth, A., Smets, B., d'Oreye, N. and Kervyn, F., 2023. Detecting sources of shallow tremor at neighboring volcanoes in the Virunga Volcanic Province using seismic amplitude ratio analysis (SARA). Bulletin of Volcanology, 85(5), p.27.

[55]Yates*, A., **Caudron, C**., Lesage, P., Mordret, A., Lecocq, T. and Soubestre, J., 2023. Assessing similarity in continuous seismic cross-correlation functions using hierarchical clustering: application to Ruapehu and Piton de la Fournaise volcanoes. Geophysical Journal International, 233(1), pp.472-489.

2022

[54]**Caudron, C.**, Aoki, Y., Lecocq, T., De Plaen, R., Soubestre, J., Mordret, A., Seydoux, L. and Terakawa, T., 2022. Hidden pressurized fluids prior to the 2014 phreatic eruption at Mt Ontake. Nature Communications, 13(1), pp.1-9.

- ➔ Editor's highlight: https://www.nature.com/collections/eihfbddfac

[53]Montanaro, C., Mick, E., Salas-Navarro, J., **Caudron, C.**, Cronin, S.J., de Moor, J.M., Scheu, B., Stix, J. and Strehlow, K., 2022. Phreatic and Hydrothermal Eruptions: From Overlooked to Looking Over. Bulletin of Volcanology, 84(6), pp.1-16.

[52]Matoza, R. S., Fee, D., Assink, J. D., Iezzi, A. M., Green, D. N., Kim, K., **C. Caudron**,... & Wilson, D. C. 2022. Atmospheric waves and global seismoacoustic observations of the January 2022 Hunga eruption, Tonga. Science, 377(6601), pp.95-100.

[51]Smittarello, D., Smets, B., Barrière, J., Michellier, C., Oth, A., Shreve, T., **C. Caudron** ... & Syavulisembo Muhindo, A. (2022). Precursor-free eruption triggered by edifice rupture at Nyiragongo volcano. Nature, 609(7925), 83-88.

[50]**Caudron, C.**, Vandemeulebrouck, J. and Sohn, R.A., 2022. Turbulence-induced bubble nucleation in hydrothermal fluids beneath Yellowstone Lake. *Communications Earth & Environment*, *3*(1), pp.1-6.

[49]**Caudron, C.,** Soubestre, J., Lecocq, T., White, R.S., Brandsdóttir, B. and Krischer, L., 2022. Insights into the dynamics of the 2010 Eyjafjallajökull eruption using seismic interferometry and network covariance matrix analyses. Earth and Planetary Science Letters, 585, p.117502.

[48]Ardid, A., Dempsey, D., **Caudron, C**., Cronin, S. Seismic precursors to the Whakaari 2019 phreatic eruption are transferable to other eruptions and volcanoes. *Nat Commun* **13,** 2002 (2022). https://doi.org/10.1038/s41467-022-29681-y2021

2021

[47]**Caudron, C.,** Girona, T., Jolly, A., Christenson, B., Savage, M.K., Carniel, R., Lecocq, T., Kennedy, B., Lokmer, I., Yates, A. and Hamling, I., 2021. A quest for unrest in multiparameter observations at Whakaari/White Island volcano, New Zealand 2007–2018. Earth, Planets and Space, 73(1), pp.1-21. (Invited Frontiers Letter)

[46]Nowé Sylvain, Thomas Lecocq, **Corentin Caudron**, Kristín Jónsdóttir and Frank Pattyn "Permanent, seasonal and episodic seismic sources around Vatnajokull, Iceland from the analysis of correlograms", Volcanica, 4(2).

[45]Cubuk-Sabuncu, Y., Jónsdóttir, K., **Caudron, C**., Lecocq, T., Parks, M.M., Geirsson, H. and Mordret, A., Temporal Seismic Velocity Changes During the 2020 Rapid Inflation at Mt. Þorbjörn-Svartsengi, Iceland, Using Seismic Ambient Noise. *Geophysical Research Letters*, p.e2020GL092265.

[44]Bernard, A., Villacorte, E., Maussen, K., **Caudron, C**., Robic, J., Maximo, R.P., Rebadulla, R., Bornas, M., Solidum Jr., R., 2021, Carbon dioxide in Taal volcanic lake: a simple gasometer for volcano monitoring, Geophysical Research Letters, 47, e2020GL090884

- ➔ Highlight in Nature Reviews Earth Environment: Scott, E. Volcanic Gasometer. Nat Rev Earth Environ (2020). https://doi.org/10.1038/s43017-020-00135-7

2020

[43]Lecocq, T., Hicks, S.P., Van Noten, K., van Wijk, K., Koelemeijer, P., De Plaen, R.S., Massin, F., Hillers, G.,…, **Caudron C.**,…, 2020. Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures. Science.

[42]Inguaggiato C., Pappaterra S., Peiffer, L., Apollaro C., Brusca L., De Rosa R., Rouwet D., **Caudron C**., Suparjan, 'Mobility of REE from hyperacid brine to secondary minerals precipitated in volcanic hydrothermal systems: Kawah Ijen volcanic lake (Indonesia)', Science of The Total Environment, 140133.

[41] **C. Caudron**, M. De Batist, G. Jouve, G. Matte, T. Hermans, A. Flores-Orozco, W. Versteeg, Z. Ghazoui, P. Roux, J. Vandemeulebrouck, and B. Messages in the Bubbles, EOS.

[40] Ashton F. Flinders, **Caudron, C.** Ingrid A. Johanson, Taka'aki Taira, Brian Shiro, Matthew Haney, Seismic velocity variations associated with the 2018 Lower East Rift Zone eruption of Kīlauea, Hawai'i, Special Volume on the 2018 Kīlauea eruption, Hawai'i, Bulletin of Volcanology, 82, p.47. IF2016: 2.58.

[39]Anna B Perttu, **Caudron, C.**, Jelle D Assink, Dirk Metz, Dorianne Tailpied, Brian Perttu, Clément Hibert, David Fee, Dini Nurfiani, M. Muzli, Øystein L Andersen, Benoit Taisne. Reconstruction of the 2018 Tsunamigenic Flank Collapse and Eruptive Activity at Anak Krakatau Based on Eyewitness Reports, Seismo-acoustic and Satellite Observations, Earth and Planetary Science Letters. IF2016: 4.409.

[38]Jolly, A., **Caudron, C.**, Girona, T., Christenson, B., Carniel, R. 'Silent' Dome Emplacement into a Wet Volcano: Observations from an Effusive Eruption at White Island (Whakaari), New Zealand in Late 2012. Geosciences 2020, 10, 142, Special Issue, Exploring and Modeling the Magma-Hydrothermal Regime, Geosciences 10, no. 4 (2020): 142. IF: 1.82

[37]Métaxian, J. P., Santoso, A. B., **Caudron, C.**, Cholik, N., Labonne, C., Poiata, Beauducel, F., Monteiller, V., Fahmi, A.A., Rizal, M.H., Nandaka, I.M.A, Migration of seismic activity associated with phreatic eruption at Merapi volcano, Indonesia, Journal of Volcanology and Geothermal Research, 106795. IF_{2016}: 2.492.

[36]**Caudron, C.**, Chardot, L., Girona, T., Aoki, Y., & Fournier, N. Towards Improved Forecasting of Volcanic Eruptions: editorial, Frontiers in Earth Science, 8, 45. IF_{2019}: 2.892.

2019

[35]*Donaldson, C., T. Winder, **C. Caudron**, R.S. White, Crustal seismic velocity responds to seasonal loading and a magmatic intrusion in Iceland's Northern Volcanic Zone, Science Advances, 5 (11). IF_{2017}: 11.51.

[34]Cassidy, M., S. K. Ebmeier, C. Helo, S. F. L. Watt, **C. Caudron**, A. Odell, K. Spaans et al. "Explosive eruptions with little warning: Experimental petrology and volcano monitoring observations from the 2014 eruption of Kelud, Indonesia." Geochemistry, Geophysics, Geosystems, 20 (8), 4218-4247 IF_{2016}: 3.201.

- ➔ Research Spotlight in EOS:
  "Explosive Volcanic Eruption Powered by Water-Saturated Magma"
- ➔ among the top 10% most downloaded papers in Geochemistry, Geophysics, Geosystems

[33]**Corentin Caudron**, Társilo Girona, Benoît Taisne, Suparjan, Hendra Gunawan, Kristianto, Kasbani, Change in seismic attenuation as a long-term precursor of gas-driven eruptions, Geology, 10.1130/G46107.1, 47, 7, (632-636). IF_{2018}: 5.073.

[32]T. Girona, **Caudron, C.**, C. Huber, Origin of the shallow seismicity preceding volcanic eruptions, Origin of shallow volcanic tremor: The dynamics of gas pockets trapped beneath thin permeable media. Journal of Geophysical Research, 124, 4831-4861. IF_{2016}: 3.201.

[31]*De Plaen S.M., Cannata A., **Caudron C.,** Lecocq T., Cannavo F. Temporal changes of seismic velocity caused by volcanic activity at Mt. Etna revealed by the autocorrelation of ambient seismic noise. Special Issue of Frontiers in Earth Science 'Towards improved forecasting of volcanic eruptions', 6, 251. IF_{2019}: 2.892.

[30]*Yates A., Savage, M.K., Jolly A.D., **Caudron C.**, Hamling I.J. Volcanic, co-seismic, and seasonal changes detected at White Island (Whakaari) volcano, New Zealand, using seismic ambient noise. Geophysical Research Letters, 46, 99-108. IF_{2016}: 4.253.

- ➔ among the top 10% most downloaded papers in Geophysical Research Letters

[29]Taisne B., Perttu A., Tailped D., **Caudron C.**, Le Pichon A. Atmospheric control on ground and space based early warning system for hazard linked to ash injection into the atmosphere. Book Chapter on Infrasound Monitoring (Challenges and New Perspectives)

2018

[28]**Caudron C.**, B. Taisne, J. Neuberg, A. Jolly, B. Christenson, T. Lecocq, Suparjan, D. Syahbana, G. Suantika. Anatomy of a hidden phreatic eruption. Earth, Planets and Space, 70:168, Special Issue in Earth, Planets and Space "Towards forecasting phreatic eruptions: Examples from Hakone volcano and some global equivalents", 70:168.

[27]Byrdina, S., Grandis, H., Sumintadireja, P., **Caudron, C.**, Syahbana, D.K., Naffrechoux, E., Gunawan, H., Suantika, G. and Vandemeulebrouck, J., 2018. Structure of the acid hydrothermal system of Papandayan volcano, Indonesia, investigated by geophysical methods. Journal of Volcanology and Geothermal Research, 358, pp.77-86.

[26]**Caudron C.**, Bernard A., Murphy S., Inguaggiato S., Gunawan H. Volcano-hydrothermal system and activity of Sirung volcano (Pantar Island, Indonesia), Journal of Volcanology and Geothermal Research, 357, 186-199.

[25]Woods J., *Donaldson, C., White R.S., **Caudron C.**, Brandsdóttir B., Hudson T.S., Ágústsdóttir T., Long-period seismicity reveals magma pathways above a laterally propagating dyke during the 2014-15 Bárðarbunga rifting event, Iceland, Earth Planetary Science Letters, 490, 216-229.

[24]**Caudron C.**, R.W. White, R.G. Green, J. Woods, T. Ágústsdóttir, *C. Donaldson, T. Greenfield, E. Rivalta, B. Brandsdóttir, 2018. Seismic Amplitude Ratio Analysis of the Bárdabunga-Holuhraun dike propagation and eruption, Journal of Geophysical Research: Solid Earth, 123(1), pp.264-276.

- ➔ Research Spotlight in EOS:
  "A novel way to track magma flow"
- ➔ Among the top 10% most downloaded papers in Journal of Geophysical Research: Solid Earth

[23]T. Girona, C. Huber, **Caudron, C.**, 2018. Sensitivity to lunar cycles prior to the 2007 phreatic eruption of Ruapehu volcano, Nature Scientific Reports, 8, no. 1: 1476 IF_{2016}: 4.259

- ➔ Top 100 in Earth Science (in 2019)

2017

[22]*Donaldson C., **Caudron C.**, Green R.G., Thelen W.A., White R.S., 2017. Relative seismic velocity variations correlate with deformation at Kilauea volcano, Science Advances 3 (6). IF_{2017}: 11.51.

[21]**Caudron C.**, Campion R., Rouwet D., Capaccioni B., Lecocq T., Syahbana D.K., 2017. Stratification at the Earth's largest acidic lake and its consequences, Earth and Planetary Science Letters, 459, 28-35. IF_{2016}: 4.409.

[20]**Caudron C.**, Ohba T., Capaccioni B., 2017. Geochemistry and Geophysics of volcanic lakes: an introduction, Geological Society of London, SP437 (1), 1-8. IF_{2016}: 3.037.

[19]Perton M., *Spica Z., **Caudron C.** , 2017. Inversion of the horizontal to vertical spectral ratio in presence of strong lateral heterogeneity, Geophysical Journal International. IF_{2016}: 2.414.

[18]Van Camp M., de Viron O., Meurers B., Francis O., Watlet A., **Caudron C.**, 2017. Geophysics From Terrestrial Time-Variable Gravity Measurements, Reviews of Geophysics 55, no. 4, 938-992. IF_{2016}: 12.34.

2016

[17]**Caudron C.**, Taisne B., Perttu A., Garcés M., Silber E.A., Mialle P., 2016. Infrasound and seismic detections associated with the 7 September 2015 Bangkok fireball. Geoscience Letters, 3(1), p.26. IF_{2016}: 2.243.

[16]*De Plaen R.S., Lecocq T., **Caudron C.**, Ferrazzini V., Francis O., 2016. Single-station monitoring of volcanoes using seismic ambient noise. Geophysical Research Letters, 43(16), pp. 8511-8518. IF_{2016}: 4.253.

[15]Lewicki J.L., **Caudron C.,** van Hinsberg V.J., Hilley G.E., 2016. High spatio-temporal resolution observations of crater lake temperatures at Kawah Ijen volcano, East Java, Indonesia. Bulletin of Volcanology, 78(8), pp. 1-11. IF_{2016}: 2.58.

[14]Tonini R., Sandri L., Rouwet D., **Caudron C.**, Marzocchi W., Suparjan. 2016. A new Bayesian Event Tree tool to track and quantify volcanic unrest and its application to Kawah Ijen volcano. Geochemistry, Geophysics, Geosystems,17(7), pp. 2539-2555. IF_{2016}: 3.201.

- ➔ Research spotlight in EOS:
  "A new tool to better forecast volcanic unrest" S. Hall, vol.97, 2016-07-08.

[13]Gunawan H., **Caudron C.**, Pallister J. et al., New insights into Kawah Ijen hydrothermal system from the wet volcano workshop experiment, Geological Society of London, Special Publications, 2016, In: Ohba, T., Capaccioni, B., Caudron, C. (eds) Geochemistry and Geophysics of Active Volcanic Lakes, SP437. IF_{2016}: 3.037.

[12]**Caudron C.**, Mauri G., Williams-Jones G., Lecocq T., Syahbana D.K., De Plaen R., Peiffer L., Bernard A., Saracco G. New insights into the Kawah Ijen hydrothermal system from

geophysical data, Geological Society of London, Special Publications, 2016, In: Ohba, T., Capaccioni, B., Caudron, C. (eds) Geochemistry and Geophysics of Active Volcanic Lakes, SP437. IF_{2016}: 3.037.

2015

[11]**Caudron C.**, Taisne B., Kugaenko Y., Saltykov V. ,2015. Magma migration at the onset of the 2012-13 Tolbachik eruption revealed by Seismic Amplitude Ratio Analyses. Journal of Volcanology and Geothermal Research, Special issue on Tolbachik 2012-2013 eruption, 307, 60-67. IF_{2016}: 2.492.

[10]**Caudron C.**, Taisne, B., Garces, M., Le Pichon, A, Mialle, P. ,2015. On the use of remote infrasound and seismic stations to constrain eruptive sequence and intensity during the 2014 Kelud eruption. Geophysical Research Letters, 42, 6614-6621, doi: 10.1002/2015GL064885. IF_{2016}: 4.253.

[9]**Caudron C.**, Lecocq T., Syahbana D.K., Watlet A., Camelbeeck T., Bernard A., S. Surono. ,2015. Stress and mass changes during the 2011-2012 volcanic unrest at Kawah Ijen volcano (East Java, Indonesia). Journal of Geophysical Research, 120, 5117-5134. IF_{2016}: 3.318.

[8]*Spica Z., **Caudron C.,** Lecocq T., Perton M., Piña-Flores J., Camelbeeck T., Legrand D., Syahbana D.,2015. Velocity models and site effect at Kawah Ijen volcano and Ijen caldera (Indonesia) from correlation of ambient noise. Journal of Volcanology and Geothermal Research, 302, 173-189, doi:10.1016/j.jvolgeores.2015.06.016. IF_{2016}: 2.492.

[7]**Caudron C.,** Syahbana D.K., Lecocq T., Van Hinsberg V., McCausland W., Triantafyllou A., Camelbeeck T., Bernard A., Surono.,2015. Kawah Ijen volcanic activity: a review. Bulletin of Volcanology, 77(3), 39 pp. IF_{2016}: 2.58.

[6]Krischer L., Megies T., Barsch R., Beyreuther M., Lecocq T., **Caudron C.,** Wassermann J. ObsPy: A Bridge for Seismology into the Scientific Python Ecosystem. Computational Science & Discovery 8 014003, doi:10.1088/1749-4699/8/1/014003.

2014

[5]Gunawan H., Pallister J., **Caudron C.** Multidisciplinary Monitoring Experiments at Kawah Ijen Volcano, Eos Trans. AGU, 95(48), 447, doi: 10.1002/2014EO480003.

[4]Syahbana D., **Caudron C.**, Lecocq T., Jousset P., Camelbeeck T., Bernard A., Surono S., 2014. Fluid dynamics inside a 'wet' volcano inferred from the complex frequencies of long-period (LP) events: An example from Papandayan volcano, West Java, Indonesia, during the 2011 seismic unrest., Journal of Volcanology and Geothermal Research, Vol. 280, 76-89, doi: 10.1016/j.jvolgeores.2014.05.005. IF_{2016}: 2.492.

[3]Lecocq, T., **Caudron C.**, F. Brenguier, 2014. MSNoise: a Python package for computing and monitoring seismic velocity changes using ambient seismic noise, Seismological Research Letters, vol. 85, 715-726, doi:10.1785/0220130073. IF_{2016}: 3.275.

2013

Caudron C., Multi-disciplinary continuous monitoring of Kawah Ijen volcano, East Java, Indonesia, PhD Thesis, ULB, Brussels, Belgium.

2012

[2]Campion R., Martinez-Cruz M., Lecocq T., **Caudron C.**, Pacheco J., Pinardi G., Hermans C., Carn S., Bernard A. ,2012. Space- and ground-based measurements of sulphur dioxide emissions from Turrialba Volcano (Costa Rica). Bulletin of Volcanology, vol. 74, 1757–1770, doi:10.1007/s00445-012-0631-z. IF_{2016}: 2.58.

[1]**Caudron C.**, Mazot A., Bernard A. ,2012. Carbon dioxide dynamics in Kelud volcanic lake. Journal of Geophysical Research, vol. 117, doi:10.1029/2011JB008806. IF_{2016}: 3.318.

Non-peer-reviewed

T. Camelbeeck, Lecocq, T., Syahbana, D., **Caudron C.**, Surono Surono (2014). Cooperation in Volcano-Seismology between the Royal Observatory of Belgium and the Indonesia Center for Volcanology and Geological Hazard Mitigation, Bulletin des Séances de l'Académie Royale des Sciences d'Outre-Mer. 01/2013; 59(2-4):407-417.

Conference abstracts cited

Jouve, G., **Caudron, C**. and Matte, G., 2021, September. Gas detection and quantification using iXblue Echoes high-resolution sub-bottom profiler and Seapix 3D multibeam echosounder from the Laacher See (Eifel, Germany). In *OCEANS 2021: San Diego–Porto* (pp. 1-4). IEEE.

Caudron C., Bernard, A. Hydroacoustic quantifications of CO_2 bubbles in volcanic lakes. 7th Workshop on Volcanic Lakes, 2010, Costa Rica.

Caudron C., Lecocq T., Syahbana D.K., Camelbeeck T., Bernard A., Surono. Multi-disciplinary continuous monitoring of Kawah Ijen volcano, East Java, Indonesia. AGU Fall Meeting 2012, San Francisco, USA.

Report

SAC29 Report Montserrat volcano 2024

INVITED SEMINARS/TALKS

(1) Les lacs volcaniques pour appréhender les comportements magmatiques/hydrothermaux des volcans
Société Royale Belge d'Astronomie, de Météorologie et de Physique du Globe, Uccle, Belgium, May 2011

(2) Carbon dioxide dynamics in Kelud volcanic lake
ISTerre, Chambéry, France, March 2013

(3) Multi-disciplinary continuous monitoring of Kawah Ijen volcano, East Java, Indonesia
MRAC, November 2013, Brussels, Belgium

(4) Monitoring at Mayon volcano: recent results and perspectives
Phivolcs, Philippines, March 2014

(5) Seismic and infrasound monitoring of volcanoes in South East Asia
ISTerre, Chambéry, France June 2015

(6) Monitoring volcanic eruptions in South East Asia using infrasound
CEA, Paris, France, June 2015

(7) Infrasound and seismic monitoring of volcanoes in South East Asia
VUB, Brussels, Belgium, October 2015

(8) Monitoring South East Asian volcanoes using infrasounds
University of Cambridge, UK, January 2016

(9) Infrasound and seismic monitoring of volcanoes in South East Asia

University of Leeds, UK, March 2016
(10) Insights into wet volcanoes using multi-disciplinary monitoring
University of Bologna, March 2017
(11) Infrasound monitoring of volcanoes
MRAC, Tervuren, Belgium, March 2017
(12) Vers une meilleure compréhension des volcans 'mouillés'
ISTerre, Chambéry, France, February 2018
(13) Insights into 'wet volcanoes' from multi-disciplinary approaches
Ghent University, Belgium, February 2018
(14) Dynamics of volcano-hydrothermal systems
EOST Strasbourg, France, January 2019
(15) Towards forecasting gas-driven eruptions
EOS Singapore, March 2019
(16) Towards real-time monitoring of volcanoes using seismic noise
3rd Southeast Asian Conferences on Geophysics (SEACG), ITB, Bandung, Indonesia, November 2020
(17) Improved understanding of volcanoes using seismic noise
100th Anniversary of Volcano Monitoring in Indonesia, CVGHM, Bandung, Indonesia, November 2020
(18) Vers une meilleure compréhension des systèmes volcano-hydrothermaux
Académie Royale de Sciences d'Outre-Mer, Belgium, November 2020
(19) Improved understanding of volcanoes using continuous seismic recordings
ISTerre, France, January 2021
(20) Improved understanding of volcanoes using continuous seismic recordings
Potsdam, Germany, June 2021
(21) Vers une surveillance du dégazage volcanique par méthodes sismo-acoustiques, iXtech 2021,
Saint-Germain en Laye, France, July 2021
(22) The Past and Future Volcanic Activity on Kawah Ijen
International volcano academy - Ijen Geopark (virtual, December 2021)
(23) Hydroacoustique du bullage volcanique
Thonons-les-Bains INRAE (France, Mars 2022)
(24) Towards monitoring volcanoes using continuous-based seismicity
Belgian National Committee for Geodesy and Geophysics (IUGG Belgium, June 2022)
(25) Monitoring of calderas using Distributed Acoustic Sensing
OV Monitoring webinar (Belgium, November 2022)
(26) Monitoring degassing using Fiber Optic cables
Tips Seminar, 2024
(27) Corentin Caudron, Alexander Yates, Jean Soubestre and Thomas Lecocq
Towards monitoring volcanoes using continuous-based seismicity
Invited at AGU 2021 (USA)
(28) Towards monitoring phreatic eruptions using seismic noise
Corentin Caudron, Társilo Girona, Thomas Lecocq, Alberto Ardid, David Dempsey, and Alexander Yates, IAVCEI meeting 2023 (New Zealand), Invited
(29) Towards monitoring phreatic eruptions using seismic noise
Corentin Caudron, Társilo Girona, Thomas Lecocq, Alberto Ardid, David Dempsey, and Alexander Yates, EGU meeting 2023 (Austria), Invited
(30) Towards monitoring volcanic hydrothermal alteration using geophysical approaches
Corentin Caudron, Thomas Lecocq, Alexander Yates, David Dempsey, Alberto Ardid, Thomas Hermans, Lore Vanhooren, Olivier Fontaine, Tom Bultreys, and Társilo Girona
EGU meeting 2023 (Austria), Invited
(31) Towards monitoring phreatic eruptions
Corentin Caudron
Evaluation of the state of Campi Flegrei, chance of any phreatic events and possible mitigation, Invited, March 2024
(32) Monitoring volcanoes using fiber optic cables

Corentin Caudron and Ben Roche
ULB Polytech fiber group, April 2024

(33) Towards monitoring volcanoes using continuous-based seismicity
Corentin Caudron
ITB Bandung, Indonesia, October 2024

(34) Towards monitoring volcanoes using continuous-based seismicity
Corentin Caudron
GEO@EAIFR Webinar Series 2024, Kigali, Rwanda, August 2024

(35) Volcano-hydrothermal systems and lakes
Corentin Caudron
IGCP 767 Project; Unesco Symposium: Western Rwanda and its environmental challenges, Kigali, Rwanda, November 2024

(36) Towards monitoring volcanoes using fibers
Corentin Caudron
Fluves seminar, Ghent, December 2024

(37) Vers une prévision des éruptions volcaniques
Corentin Caudron
Cepulb seminar, Brussels, December 2024

(38) Vers une prévision des éruptions volcaniques
Corentin Caudron
STEMentiel seminar, Brussels, May 2025

(39) Towards forecasting volcanic eruptions using passive vibrations
Corentin Caudron
LMU Munich seminar, Germany, May 2025

WORKSHOP organizer-presenter

- IAVCEI 2017, Kawah Ijen workshop with K. Berlo (August 2017, Portland, USA, ~15 participants)
- IAVCEI 2017, MSNoise: From Noise to Data with T. Lecocq (August 2017, Portland, USA, ~20 participants)
- TSG-VMSG-BGA Liverpool 2017, NERC Advanced Training Workshop: Python solutions for management of continuous seismic data and ambient noise interferometry with T. Lecocq, S. De Angelis (January 2017, Liverpool, UK, ~30 participants)
- TIDES (MS)Noise Workshop Vienna 2016 with T. Lecocq, A. Mordret (April 2016, Vienna, Austria, 31 participants)
- EOS 2015 workshop: MSNoise workshop (October 2015, Singapore, Singapore, ~15 participants).
- EOS 2015 workshop: MSNoise workshop (May 2015, Singapore, Singapore, ~15 participants).
- AGU 2014 related workshop: MSNoise workshop: with T. Lecocq, F. Brenguier (December 2014, San Francisco, USA, 13 participants).
- COV 2014 post-workshop: “Wet volcano workshop” with J. Pallister, H. Gunawan, B. Christenson, S. Primulyana (September 2014, Kawah Ijen, Indonesia, ~25 participants)
- AGU 2013 related workshop: MSNoise/MFAST-Tessa to monitor changes in isotropic and anisotropic seismic velocity: with T. Lecocq, F. Brenguier (December 2013, San Francisco, USA, ~30 participants).
- IAVCEI 2013 pre-workshop: “MSNoise: a Python Package for Monitoring Seismic Velocity Changes using Ambient Seismic Noise” in “New methods/computer codes for volcano monitoring” with T. Lecocq, F. Brenguier (July 2013, Kagoshima, Japan, ~15 participants)
- CVGHM Bandung 2012 workshop: Volcano seismic monitoring: data analysis with Python with T. Lecocq (July 2012, Bandung, Indonesia, ~15 participants)